(FS) 1. Continuous random variables HW

NO. 1

The continuous random variable X has cumulative distribution function F given by

$$\mathrm{F}(x) = \begin{cases} 0 & x < 0, \\ \frac{1}{60}(16x - x^2) & 0 \leqslant x \leqslant 6, \\ 1 & x > 6. \end{cases}$$

a. Find the interquartile range of X. [4]

b. Find $\mathrm{E}(X^3)$. [4]

The random variable Y is such that $Y = \sqrt{X}$.

c. Find the probability density function of Y. [3]

NO. 2

The continuous random variable X has probability density function f given by

$$\mathrm{f}(x) = \begin{cases} \frac{1}{5}x & 0 \leqslant x < 2, \\ \frac{2}{15}(5-x) & 2 \leqslant x \leqslant 5, \\ 0 & \text{otherwise.} \end{cases}$$

a. Find the cumulative distribution function of X. [3]

b. Find the median value of X. [2]

c. Find $\mathrm{E}(X^2)$. [2]

d. Find $\mathrm{P}(1 \leq X \leq 3$. [2]

NO. 3

The continuous random variable X has distribution function given by

$$F(x) = \begin{cases} 0 & x < 0, \\ \frac{1}{90}(x^2 + x^4) & 0 \leqslant x \leqslant 3, \\ 1 & x > 3. \end{cases}$$

The random variable Y is defined by $Y = X^2$.

i. Find the probability density function of Y. [4]

.....

ii. Find the mean value of Y. [2]

.....

NO. 4

The continuous random variable X has probability density function f given by

$$\mathrm{f}(x) = \begin{cases} \frac{1}{6}\left(x^{-\frac{1}{3}} - x^{-\frac{2}{3}}\right) & 1 \leqslant x \leqslant 27, \\ 0 & \text{otherwise}. \end{cases}$$

a. Find the cumulative distribution function of X. [3]

The random variable Y is defined by $Y = X^{\frac{1}{3}}$.

b. Find the probability density function of Y. [3]

c. Find the exact value of the median of Y. [2]

NO. 5

The continuous random variable X has cumulative distribution function F given by

$$\mathrm{F}(x) = \begin{cases} 0 & x < 0, \\ 1 - \frac{1}{144}(12-x)^2 & 0 \leqslant x \leqslant 12, \\ 1 & x > 12. \end{cases}$$

a. Find the upper quartile of X. [2]

b. Find Var(X^2). [5]

The random variable Y is given by $Y = \sqrt{X}$.

c. Find the probability density function of Y. [3]

NO. 6

The continuous random variable X has cumulative distribution function F given by

$$\mathrm{F}(x) = \begin{cases} 0 & x < -1, \\ \frac{1}{2}(1+x)^2 & -1 \leqslant x \leqslant 0, \\ 1 - \frac{1}{2}(1-x)^2 & 0 < x \leqslant 1, \\ 1 & x > 1. \end{cases}$$

a. Find the probability density function of X. [2]

b. Find $\mathrm{P}\left(-\frac{1}{2} \leqslant X \leqslant \frac{1}{2}\right)$. [2]

c. Find $\mathrm{E}(X^2)$. [2]

d. Find $\mathrm{Var}(X^2)$. [2]